

UDEE Practice Test 07: Answer Key

30 questions · Section A Figural, Section B Observation (interference), Section C Logic-Based Reasoning · Printed separately from the test sheet

Review rules. Read the explanation for every question you missed AND every question you guessed correctly. Pay extra attention to Q25 (contrapositive of a positive conditional) and Q28 (denying the antecedent, a new trap not in tests 05 or 06).

Answer Grid

1. A	2. D	3. B	4. B	5. C
6. E	7. D	8. A	9. C	10. B
11. B	12. C	13. B	14. A	15. B
16. B	17. B	18. B	19. B	20. C
21. C	22. A	23. C	24. B	25. B
26. B	27. D	28. D	29. D	30. B

Section A: Figural Reasoning

Q1: A. Arrow left, dot on the RIGHT side

Reflection across a vertical axis flips the figure left-to-right. The arrow tip swings from RIGHT to LEFT. The dot, originally on the LEFT side of the arrow, now sits on the RIGHT side. Option B puts the dot on the wrong side. C is unchanged. D rotates instead of reflecting. E places the dot in the middle instead of mirroring its position.

Q2: D. 18 rectangles

For an m-by-n grid, the number of rectangles equals $C(m+1, 2) \times C(n+1, 2)$. A 2-row by 3-column grid has 3 horizontal lines and 4 vertical lines, so: $C(3, 2) \times C(4, 2) = 3 \times 6 = 18$. Option A (6) counts only single cells. B (9) and C (12) are partial counts. E (24) overcounts.

Q3: B. Blue arrow pointing DOWN

Rows rotate the arrow: row 1 up (0 degrees), row 2 right (90 CW), row 3 down (180). Columns change color: col 1 red, col 2 green, col 3 blue. Row 3, column 3 therefore shows a BLUE arrow pointing DOWN. Option A keeps the row 1 rotation. C and D keep the wrong color. E takes the row 2 rotation.

Q4: B. 4 space diagonals

A cube has 8 vertices. Each vertex has exactly one vertex that does NOT share a face with it: the vertex diagonally opposite through the cube's interior. Pairing the 8 vertices into opposite pairs gives $8 / 2 = 4$ space diagonals. Option C (6) is the number of faces, not the number of space diagonals. E (12) confuses space diagonals with edges.

Q5: C. 34

The sequence is Fibonacci-style: each term equals the sum of the previous two. Check: $3 + 5 = 8$, $5 + 8 = 13$, $8 + 13 = 21$, $13 + 21 = 34$. Option A (28) adds $21 + 7$. B (30) is $21 + 9$. D (42) doubles 21. E (55) takes one term too far (next after 34).

Q6: E. The plus-pentomino

Figures A through D are all T-pentominoes (3 cells horizontal with a 2-cell stem from the middle) in the four possible planar rotations. Figure E is the plus-pentomino (5 cells arranged as a plus sign), a completely different piece that is NOT a rotation of the T. The plus-pentomino has 4-fold symmetry (looks the same from every 90 degree rotation), while the T has only 1-fold symmetry.

Q7: D. 8 holes

Two folds in half create $2^2 = 4$ layers. Each punched hole passes through all 4 layers, creating 4 holes on the unfolded sheet. Two separate holes therefore create $2 \times 4 = 8$ total holes. Option A ignores the layering. B counts only one punch's worth. E (16) over-counts by doubling incorrectly.

Q8: A. Three pairwise lenses shaded, center triple region empty

"Inside exactly two of the three circles" means the three pairwise overlap lenses MINUS the central triple overlap. The shaded region forms three crescent-like slivers that meet at, but do not include, the center triangular triple-overlap. Option B shades only the center (inside all three). Option C shades the full union. Option D shades the outer single-circle moons (inside exactly one). None of those match "exactly two."

Q9: C. Southeast

Rotating 45 degrees COUNTER-CLOCKWISE from North cycles: N to NW to W to SW to S to SE to E to NE to N. Five steps from N: step 1 NW, step 2 W, step 3 SW, step 4 S, step 5 SE. Formally: $(5 \times 45) = 225 \text{ degrees CCW from N} = \text{SE}$. Trap answers A (NE) and E (W) are different step counts from the same rule.

Q10: B. Larger than step 4, pointing DOWN

Two simultaneous rules: the arrow grows each step, and the direction rotates 45 degrees clockwise each step. Step 1 UP, step 2 NE (45 CW), step 3 E (90 CW), step 4 SE (135 CW), step 5 S = DOWN (180 CW). Size continues to grow, so step 5 is LARGER than step 4. Option A keeps the same size. C advances direction wrongly. D shrinks. E fails both rules.

Section B: Observation with Interference

Q11: B. Capital Bowl 2026

The red banner across the upper stadium bowl read "CAPITAL BOWL 2026" in white lettering. Option C ("Capitol Games") is the government-building trap; D substitutes "Classic" for "Bowl"; E is a throwaway. Option A is the most plausible misread.

Q12: C. 18:47

The digital clock mounted on the stadium wall read **18:47** in red LED. Scene 14's clock read 2:30 PM (analog). Do not let Scene 14's time overwrite this one.

Q13: B. 2 gates

Two metal detector gates were visible: labeled GATE 1 (left) and GATE 2 (right), each with vertical pillar frames.

Q14: A. A bomb-sniffing dog

A K-9 unit with a handler in dark uniform stood between the two metal detector gates, with the dog on a leash to the right of the handler. No mounted unit, drone operator, or SWAT team was shown.

Q15: B. Bright yellow

The security guard between the two gates wore a bright yellow high-visibility vest over dark pants. The color was unmistakable against the gray concourse. Trap A ("orange") is a common high-vis color but was not used in this scene.

Q16: B. ICU-14B

The ICU door bore the identifier "ICU-14B" in black text. Option C (ICU-41B) transposes the digits, the classic number-swap trap. Option A drops the "1" entirely.

Q17: B. 2:30 PM

The analog wall clock on the right side of the corridor showed hands at 2:30 (hour hand at 2, minute hand straight down at 6), with the label "2:30 PM" below. The 18:47 from Scene 13 is the interference.

Q18: B. 2 agents

Two Secret Service agents in dark suits with earpieces flanked the ICU door: one on the left of the door, one on the right. Both had visible earpiece lines and lapel indicators.

Q19: B. A medication cart

The nurse in the corridor (white uniform) was pushing a gray medication cart with two shelves and

visible wheels. No wheelchair, laundry bin, or food tray appeared.

Q20: C. Both scenes had clocks: Scene 13 digital (18:47), Scene 14 analog (2:30 PM)

Scene 13 showed a digital stadium clock at 18:47. Scene 14 showed an analog wall clock at 2:30 PM. Both are correctly identified in option C. Options A and B each drop one scene's clock. D mis-identifies Scene 14 as digital.

Section C: Logic-Based Reasoning

Q21: C. The incident occurred in Zone A or Zone B (or both)

"ONLY X alerts Dispatch" means $\text{alert} \rightarrow X$. Given that an alert WAS raised, by modus ponens the incident must have been of type X, which is "Zone A or Zone B." We cannot determine which specific zone, so A and B each over-specify. Option D contradicts the rule. Option E is too cautious; the "only if" relationship is strong enough to conclude zone membership.

Q22: A. T, R, P, S, Q, V, U

Verify all six constraints:

(1) P at position 3: yes.

(2) Q immediately after S: S at 4, Q at 5.

(3) T before P: T at 1, before P at 3.

(4) U at position 7: yes.

(5) R not at position 1: R at 2.

(6) V at position 6: yes.

Option B puts P at position 1. Option C puts P at position 5. Option D puts R at position 1. Option E puts Q before S.

Q23: C. 8

Pigeonhole: $\lceil 37/5 \rceil = \lceil 7.4 \rceil = 8$. Distribution of $8+8+7+7+7 = 37$ achieves max = 8. Option A (6) fails because $5 \times 6 = 30$, short by 7 cases. B (7) fails because $5 \times 7 = 35$, short by 2. Only 8 forces the total to reach 37.

Q24: B. Garcia

Test the three possibilities under "exactly ONE truthful":

Franklin truthful (Garcia and Hayes lying): Franklin says Garcia is lying, consistent. Garcia lying: "Hayes is lying" is false, so Hayes is truthful. But we said Hayes is lying. Contradiction.

Garcia truthful (Franklin and Hayes lying): Garcia says Hayes is lying, consistent. Franklin lying: "Garcia is lying" is false, so Garcia is truthful, consistent. Hayes lying: "Both Franklin and Garcia are lying" is false, so at least one of them is truthful. Garcia is truthful. All checks pass.

Hayes truthful (Franklin and Garcia lying): Hayes says both Franklin and Garcia are lying, consistent. Franklin lying: "Garcia is lying" is false, so Garcia is truthful. But we said Garcia is lying. Contradiction.

Only Garcia works as the lone truthful witness.

Q25: B. The system is NOT functioning normally

The rule is $\text{normal} \rightarrow \text{no_alert}$. An alert WAS raised means no_alert is FALSE. By modus tollens: if the consequent is false, the antecedent must be false. Therefore normal is false; the system is NOT functioning normally. Option A commits affirming the consequent in reverse. Options C and D introduce facts not in the premise. Option E is too weak; modus tollens gives a definitive answer.

Q26: B. The motion sensor was NOT triggered

The rule is $(\text{sensor} \wedge \text{unlocked}) \rightarrow \text{alarm}$. Given NOT alarm, by modus tollens: $\neg(\text{sensor} \wedge \text{unlocked}) = \neg\text{sensor} \vee \neg\text{unlocked}$. At least one of the two conjuncts is false. Given that the door WAS unlocked (so $\neg\text{unlocked}$ is false), the only remaining way to satisfy the disjunction is $\neg\text{sensor}$ being true: the motion sensor was NOT triggered. Option A contradicts the derivation. C contradicts a given fact. D and E introduce failure modes not supported by the rule.

Q27: D. 75%

Apply Bayes' theorem. Let O = overseas, A = Arabic speaker.

$$P(O) = 0.60, P(\text{domestic}) = 0.40.$$

$$P(A|O) = 0.30, P(A|\text{domestic}) = 0.15.$$

$$\text{Total: } P(A) = 0.30(0.60) + 0.15(0.40) = 0.18 + 0.06 = 0.24.$$

$$\text{Therefore } P(O|A) = P(A|O) \times P(O) / P(A) = 0.18 / 0.24 = 0.75 = 75\%.$$

Option A (18%) is $P(A \wedge O)$. Option B (30%) is $P(A|O)$. Option C (60%) is the base rate. Option E (90%) has no basis in the numbers. Only D applies Bayes correctly.

Q28: D. Nothing about Agent Woods's radio can be concluded

The rule is $L5 \rightarrow \text{radio}$. Given NOT L5. Attempting to conclude NOT radio is DENYING THE ANTECEDENT, a classic logic fallacy. Woods may still carry a radio for other reasons (personal equipment, Level 3 assignment, department issue). The rule states only that L5 is SUFFICIENT for carrying a radio, not that it is NECESSARY. Options A and B each commit this fallacy. Option E claims exclusivity ("only L5") not in the premises.

Note: Test 05 Q28 was an affirming-the-consequent trap. Test 06 Q28 was a valid forward syllogism. Test 07 Q28 is denying-the-antecedent. Three different invalid/valid patterns across the series.

Q29: D. 14:00

Anchor the knowns:

$$A = 10:00 \text{ (given).}$$

$$B = A + 2:00 = 12:00.$$

$$D = 15:00 \text{ (given).}$$

$$C = D - 1:00 = 14:00.$$

Verify C after B: 14:00 > 12:00. Consistent.

Option E (cannot determine) is the giveaway trap; the four anchors fully determine C's time.

Q30: B. S1 = true, S2 = false, S3 = true

Test the assignment:

S3 = T claims "S1 and S2 cannot both be true" = $\neg(S1 \wedge S2) = \neg(T \wedge F) = \neg F = T$. Consistent.

S2 = F claims "S1 is false." S1 is T, so the statement is false. A false statement is consistent with S2 being false.

S1 = T claims "S3 is true." S3 is T. Consistent.

All three statements align under (T, F, T).

Test the other options:

A (T, T, T): S2 claims S1 false, but S1 = T. Contradiction.

C (F, T, T): S1 = F claims S3 true; S3 = T makes the claim true, but S1 false requires its statement false. Contradiction.

D (F, F, F): S3 = F claims "S1 and S2 cannot both be true"; the statement is actually true (neither is true), so S3 should be T. Contradiction.

Only B is consistent.

Scoring: Test 07

27+

Exam-ready. You have now passed the hardest three tests in the series at strength. One more timed run if you have time, then book Pearson VUE.

23-26

Solid pass expected on the real exam. If you missed Q28 (denying the antecedent), note that this completes the trio with test-05 Q28 (affirming the consequent) and test-06 Q28 (valid syllogism). Master all three directions before exam day.

18-22

Borderline. Return to the LBR sections of tests 05, 06, and 07 side by side and drill the three syllogism directions until they are instant recognition.

<18

Stop, rebuild. You are not ready yet. Return to test 01 and work up again. Do not book Pearson VUE.

What Test 07 adds over Tests 05 and 06: Different figural patterns (mirror across vertical, rectangle counting, space diagonals of a cube, 3-circle "exactly two" region, water-table pattern growth). Scenes 13 (stadium) and 14 (hospital ICU) introduce large-crowd and medical-corridor contexts not seen in prior tests. Section C introduces Bayes probability (Q27), denying-the-antecedent trap (Q28, new), and a 4-statement self-reference (Q30 structure) whose unique consistent row is (T, F, T), different from test 05 and test 06. If you have scored 23+ across tests 05, 06, and 07, you have practiced under the highest difficulty available and are prepared.